REFLECTIVE REPORT: THE FIRST OF THEM

IMTC5002

Abstract

The reflection on the production pipeline, including pre-production, development, play testing and reflection after the game had been finished and displayed

Dan Ballem

https://github.com/DKB06060/The-First-Of-Them

1. Project Overview

1.1 Context and the RSA "Made Natural" Brief

The prototype developed for this assessment, titled The First Of Them, builds directly upon the conceptual blueprint established in the foundational pre-production planning document The First Of Them RSA Document FINALISED (1).pdf. Grounded in the Royal Society of Arts (RSA) Student Design Awards brief Made Natural, this project explores how natural systems can be reintroduced into environments where they have been displaced. The core ethos of the brief requires designers to move past anthropocentric frameworks that view nature as a resource to be dominated, controlled, extracted, or optimized. Instead, it demands innovative solutions centered on ecological repair, long-term environmental balance, and reciprocal human-nature coexistence as foundational design values.

The First Of Them explores these ideas by explicitly recontextualizing post-apocalyptic urban spaces. In mainstream interactive media—exemplified by franchises like The Last of Us (Naughty Dog, 2013-2020)—the visual motif of nature reclaiming cities is frequently used to communicate societal decay, ruin, and human vulnerability. In those titles, biological entities such as the Cordyceps infection are cast as explicitly hostile, creating an antagonistic relationship between the player and the shifting ecosystem. While highly effective for survival-horror staging and narrative tension, this framing leaves limited space to consider nature as a collaborative, restorative, or regenerative force.

This prototype deliberately subverts those conventions. Set within a familiar post-collapse world, the game shifts the player’s relationship with the environment from extraction and destruction to cultivation and systemic defence, turning natural resilience into a direct opportunity for mutual renewal.

1.2 Alignment with Sustainable Development Goal 11 (SDG 11)

Beyond the thematic prompts of the RSA brief, the prototype is structurally aligned with the United Nations’ Sustainable Development Goal 11: Sustainable Cities and Communities. Specifically, the game addresses the core targets of SDG 11.7, which mandates universal access to safe, inclusive, resilient, and green public spaces.

In traditional survival design, urban environments function as static, hostile backdrops where players scavenge, consume, and destroy to maintain individual player parameters. The First Of Them reframes urban spaces as dynamic, living systems capable of systemic recovery. By linking success to the stabilization and protection of specific urban zones, the gameplay loop demonstrates a core principle of green city theory: that urban resilience and community safety depend directly on healthy, self-sustaining ecosystems. Immersive technology is utilized here as an interactive framework to make these abstract ecological concepts tangible, operational, and directly reactive to human agency.

2. Development Process

The transition from the initial research proposal to a functional, playable prototype was scheduled across a strict 12-week solo production timeline. Because solo game development requires managing art pipelines, software engineering, level design, and system balancing simultaneously, the production workflow was split into clear thematic phases. This structured approach allowed for continuous technical evaluation and provided the structural flexibility required when handling a major mid-project architectural shift.

Weeks 1-2	Pre-Production, Research, Technical Setup
Weeks 3-5	Core Features Implementation
Weeks 6-7	Asset Production and Implementation
Weeks 8-9	Playtesting, Iteration and Addressing Feedback
Weeks 10-11	Technical Refinement and Audio Implementation
Week 12	Final Polish, Balancing and Tidying Repository Files

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2.1 Pre-Production, Research, and Technical Setup (Weeks 1-2)

The initial phase focused on moving the conceptual frameworks detailed in The First Of Them RSA Document into a concrete technical specifications list. The primary development stack was established using the Unity Engine for spatial layout and system handling, with Visual Studio serving as the Integrated Development Environment (IDE) for C# scripting. A remote repository was initialized at the start of production to ensure strict version control, continuous backup, and an audited development history, mirroring the best practices taught in lectures and observed in professional solo pipelines.

During this phase, background research was conducted into the aesthetic staging of top-down and isometric spaces. A 2D pixel art asset pipeline was selected using Aseprite. Pixel art was chosen because it provides high visual readability for tactical grids while keeping the asset production load sustainable for a solo developer within a limited timeframe. Initial block-outs of a singular urban street grid were constructed using primitive geometry to test sprite scaling, tile dimensions, and camera orthographic views before committing to final asset drawing.

2.2 The Mechanical Pivot & Core Architecture Engineering (Weeks 3-5)

Weeks 3 to 5 marked a major turning point in the development cycle. The initial proposal outlined a 2.5D action-exploration game centered on top-down combat, manual resource scavenging, a complex inventory system, and player-triggered "Green Zone" restoration nodes. However, when early prototyping began on these separate systems, the sheer volume of independent codebases required—such as character finite state machines, multi-slot inventory UIs, physics-based melee boxes, and dynamic object saving—threatened to overwhelm a solo production pipeline.

To preserve the project's thematic focus without delivering a broken, unpolished experience, a strategic design decision was made to execute a major mechanical pivot. The game was entirely restructured from an exploration-action loop into an Environmental Tower Defense Game.

In this new framework, the player acts as a strategic Firefly commander overseeing the stabilization of a specific urban zone. Instead of controlling an individual character moving through a world, the player interacts directly with the environment via a tactical grid system. To implement this system cleanly, inspiration was drawn from industry-standard object-oriented programming patterns, specifically referencing the architectural separation of data and visualization seen in SpiritWolf's Tower Defense engineering tutorials

This structural shift required writing explicit custom C# scripts to handle data management:

- TurretData.cs: A scriptable object class that stores unique values for different unit types, cost variables, and targeting ranges.
- Turret.cs: A class that uses Physics.OverlapSphere to detect enemy colliders entering a specific radius. It manages target arrays by tracking the enemy closest to the end of the path and calculates rotation angles using Quaternion.Lerp to ensure smooth visual alignment.
- WaveSpawner.cs: A timing script that handles array parsing to instantiate specific enemy prefabs at designated intervals, scaling wave difficulty based on active timer variables.

During the implementation of Turret.cs, significant technical bottlenecks arose regarding performance optimization. Running a Physics.OverlapSphere calculation inside the standard Update() loop—which executes every single frame—caused major CPU spikes and severe frame-rate drops when multiple defensive units were active on the grid. To overcome this obstacle, the code was refactored to replace frame-dependent checking with a timed C# co-routine (IEnumerator). By wrapping the proximity scanning logic in a WaitForSeconds(0.2f) loop, the target detection system was throttled to run only five times per second per unit rather than sixty. This engineering change completely stabilized the runtime performance of the prototype without sacrificing targeting accuracy.

Simultaneously, severe pathfinding logic errors emerged within the WaveSpawner.cs and enemy movement frameworks. The infected AI was programmed to iterate through an array of transform nodes designated as waypoints. However, if the vector coordinates of these path nodes were even slightly misaligned with the spatial parameters of the tile grid, the enemy colliders would fail to trigger the minimum distance threshold (Vector3.Distance). This calculation failure caused the infected AI to spin endlessly in place or completely bypass the defensive line, breaking the wave logic. To resolve this, the waypoint detection script was refactored to use an offset tolerance margin, ensuring that minor geometric imperfections in grid alignment would not stall the state progression of the incoming waves.

2.3 The Tower Defense System as an Ecological Metaphor

While moving to a tower defence framework helped control scope, it was also carefully designed to support the RSA Made Natural brief. In traditional tower defence games, military units are deployed to violently claim territory and wipe out waves of enemies. In The First Of Them, this automated loop was re-engineered to act as a direct metaphor for systemic biological feedback loops.

Real-world ecosystems do not maintain health through passive balance alone; they rely on automated checks and balances, such as apex predators regulating prey populations to prevent habitat degradation. By reframing defensive units as specialized community defence infrastructure and enemies as an aggressive, invasive fungal blight, the tactical simulation mirrors these natural biological defence mechanisms. Protecting the zone from the infected directly simulates the challenges real-world conservationists face when safeguarding fragile rewilding sites from invasive, destabilizing forces.

2.4 Asset Production & Modular Grid Integration (Weeks 6-7)

Once the underlying C# architecture was stable, production shifted to defining the game's visual presentation. The initial proposal outlined a plan to create custom, bespoke pixel art environments from scratch within Aseprite based on a 16x16 pixel dimension grid. This pipeline was designed to divide the environment into a desaturated, cold "Blighted State" representing urban neglect, and a highly saturated, warm "Reclaimed State" featuring active flora overgrowth to signal ecological recovery.

However, as development advanced, the immense programming overhead required to debug the core tower defence architecture severely compressed the time remaining for art production. Facing a rigid 12-week deadline, it became clear that drawing hundreds of individual tiles for two complete environment states would result in either an unpolished aesthetic or an unstable game build. To maintain a realistic scope, a critical executive decision was made to drop the bespoke Aseprite tile drawing pipeline and pivot to utilizing a pre-made 2D environment asset pack sourced online.

While sourcing an external asset pack resolved the immediate time crunch, it introduced significant technical friction during integration. The pre-made sprite sheets did not natively align with the custom grid spacing and sorting layers established during the Week 1 technical setup. Importing the tiles into Unity's 2D Tilemap system initially caused severe visual distortion, jagged tile clipping, and floating gaps between adjacent assets.

To resolve these integration anomalies, extensive texture re-importing and engineering calibration were required. Every sprite sheet had to be manually opened and audited to recalculate its precise Pixels-Per-Unit (PPU) setting within Unity's inspector window. Sprite pivot points had to be shifted from default centres to precise custom bottom-left anchors to ensure that placement coordinates matched the mathematical logic of the C# grid snapping scripts.

Furthermore, because the asset pack was static, the planned dynamic tile-swapping mechanic—where individual sectors would procedurally transform from dead concrete to flourishing moss upon wave completion—had to be heavily simplified. Instead of an automated, tile-by-tile transformation, the system was refactored to trigger a full-map visual overlay shift once a complete wave stage was successfully defended. This technical pivot successfully salvaged the project schedule, ensuring that visual assets were integrated cleanly without introducing engine-breaking errors, though it did require trading away a degree of granular environmental storytelling.

3. Critical Reflection

3.1 Successes and Achievements

Building The First Of Them as a solo developer within a 12-week window offered deep professional growth. It required taking full ownership of every part of the project, including technical software engineering, pixel art production, UI design, and balancing gameplay variables.

The prototype's primary success is its ability to use tactical strategy mechanics to encourage environmental reflection. Modifying an inherently combative genre like tower defence to reward systemic restoration proved that gameplay systems can communicate complex environmental ideas.

From a technical standpoint, the automated tile-swapping system works reliably. Seeing a dull, decaying urban space quickly change into a vibrant ecosystem as a direct result of tactical choices successfully delivers the core emotional premise of the original design document.

3.2 Challenges, Scope Adjustments, and Lessons Learned

The primary challenge faced during this development cycle was a profound struggle with time management and strategic planning, which forced massive, difficult scope reductions. In the pre-production phase, the programming overhead required to build a functional, stable top-down action-adventure game from scratch was severely underestimated. Even with a structured approach, writing, testing, and debugging the core C# scripts consumed a huge portion of the 12-week schedule.

As a direct consequence of technical bottlenecks, the decision was made to entirely remove the interactive Green Zone system, the scavenging/inventory mechanics, and the player-controlled combat loops from the final submission build.

Losing the player-controlled Green Zone system was the biggest compromise. The original plan for a dynamic "Regeneration Shader" that would grow moss and climbing vines across tiles in real time had to be cut to focus on ensuring the core wave logic didn't crash the engine.

Stripping out the scavenging features simplified the game's economy into basic numerical values. Originally, players were meant to gather common items to upgrade and craft new weapons, but the backend systems required to connect an inventory to the UI proved too complex to complete alongside the pathfinding logic.

Removing direct, player-controlled combat changed the game from an action hybrid into an automated tactical simulation. While this stabilized the prototype, it reduced the immediate, hands-on connection with the environment that the initial proposal aimed to create.

This experience offers a critical lesson in professional practice regarding realistic scoping. A solo developer operating on a short timeline cannot treat production like a multi-person studio pipeline.

In future projects, strict scoping frameworks like the MoSCoW method (Must have, Should have, Could have, Won't have) will be used during the initial concept phase to protect the core project goals. Cutting overambitious features early is vital; delivering a stable, functional Minimal Viable Product (MVP) is much more valuable than submitting a broken array of unpolished ideas.

4. Evaluation of Impact

Evaluating The First Of Them against the long-term goals established in Term 1 requires an honest analysis of how effectively the final tower defence prototype delivers on the requirements of the RSA brief and UN SDG 11.

4.1 Delivering on the RSA Brief

The prototype successfully addresses the central prompt of the Made Natural brief by shifting how natural systems are represented in interactive games. In traditional survival media, human development and natural systems are locked in a conflict where one must dominate the other.

By positioning the human faction (the Fireflies) as protectors who work with the ecosystem to heal a city, the project frames nature as a collaborator. The clear visual transition from ruined, grey urban structures to vibrant green zones offers a clear metaphor for urban rewilding, demonstrating how built environments can be managed to let biological life thrive alongside human communities.

3.2 Delivering on SDG 11 Goals

From an academic perspective, the core tower defence mechanics turn the urban targets of SDG 11 into clear gameplay incentives. Specifically, the loop mirrors the targets of SDG 11.7 by showing the practical value of safe, green public spaces.

In the game, neglecting a sector leaves it dangerous and prone to continuous outbreaks. Conversely, when a player successfully invests resources to clear threats and establish a green zone, that area becomes permanently safe and resilient. This directly reinforces the idea that sustainable urban care yields long-term benefits for a community's safety and well-being.

The final build proves that immersive technology can act as a safe space to explore regenerative ideas, illustrating the clear relationship between human care and environmental health.

4. Future Directions

While the current build functions effectively as a proof-of-concept prototype, there are clear paths for long-term production and potential real-world deployment.

4.1 Expanding Mechanics and Environmental Systems

The highest priority for future development is increasing the depth of both the tactical units and the ecological simulation systems. Future versions could feature multiple distinct levels, each representing a different borough of a recovering city, introducing localized environmental issues like toxic soil, concrete heat islands, or broken water drainage systems.

To counter these threats, defensive units could be expanded from basic combat roles into specialized environmental engineering roles.

4.2 Educational Use and Real-World Collaboration

Beyond commercial entertainment, The First Of Them has strong potential to serve as an educational or community engagement tool. Because the core gameplay loop prioritizes civic responsibility over resource extraction, the project is well-suited for classroom use, sustainability workshops, or public museum installations. It can act as an interactive conversation starter about what resilient, liveable cities need when facing climate change.

Furthermore, future development could be done in direct collaboration with urban rewilding organizations, environmental charities, or municipal planning boards. By integrating real-world ecological data and local plant characteristics into the game's pixel art tilesets, the prototype could let players simulate actual green infrastructure projects, bridging the gap between virtual play and meaningful community action.

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